23.1 Basic configuration

The System Tick Timer is configured using the following registers:

1. Clock Source: Select either the internal CCLK or external STCLK (P3.26) clock as the source in the STCTRL register.
2. Pins: If STCLK (P3.26) was selected as clock source enable the STCLK pin function in the PINMODE register (Section 8.5).
3. Interrupt: The System Tick Timer Interrupt is enabled in the NVIC using the appropriate Interrupt Set Enable register.

23.2 Features

- Times intervals of 10 milliseconds
- Dedicated exception vector
- Can be clocked internally by the CPU clock or by a clock input from a pin (STCLK)

23.3 Description

The System Tick Timer is an integral part of the Cortex-M3. The System Tick Timer is intended to generate a fixed 10 millisecond interrupt for use by an operating system or other system management software.

Since the System Tick Timer is a part of the Cortex-M3, it facilitates porting of software by providing a standard timer that is available on Cortex-M3 based devices.

Refer to the Cortex-M3 User Guide appended to this manual (Section 34.4.4) for details of System Tick Timer operation.

23.4 Operation

The System Tick Timer is a 24-bit timer that counts down to zero and generates an interrupt. The intent is to provide a fixed 10 millisecond time interval between interrupts. The System Tick Timer may be clocked either from the CPU clock or from the external pin STCLK. The STCLK function shares pin P3.26 with other functions, and must be selected for use as the System Tick Timer clock. In order to generate recurring interrupts at a specific interval, the STRELOAD register must be initialized with the correct value for the desired interval. A default value is provided in the STCALIB register and may be changed by software. The default value gives a 10 millisecond interrupt rate if the CPU clock is set to 100 MHz.

Remark: The maximum allowable STCLK frequency is 1/4 of CCLK.

The block diagram of the System Tick Timer is shown below in the Figure 119.

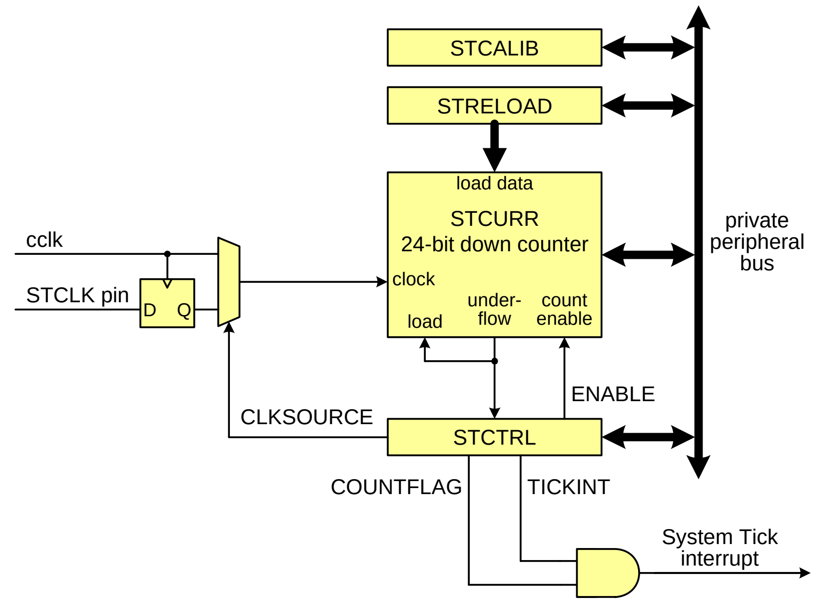

Fig 119. System Tick Timer block diagram

23.5 Register description

Table 439. System Tick Timer register map

Name	Description	Access	Reset value[1]	Address
STCTRL	System Timer Control and status register	R/W	0x4	0xE000 E010
STRELOAD	System Timer Reload value register	R/W	0	0xE000 E014
STCURR	System Timer Current value register	R/W	0	0xE000 E018
STCALIB	System Timer Calibration value register	R/W	0x000F 423F	0xE000 E01C

[1] Reset Value reflects the data stored in used bits only. It does not include content of reserved bits.

23.5.1 System Timer Control and status register (STCTRL - 0xE000 E010)

The STCTRL register contains control information for the System Tick Timer, and provides a status flag.

Table 440. System Timer Control and status register (STCTRL - 0xE000 E010) bit description

Bit	Symbol	Description	Reset value
0	ENABLE	System Tick counter enable. When 1, the counter is enabled. When 0, the counter is disabled.	0
1	TICKINT	System Tick interrupt enable. When 1, the System Tick interrupt is enabled. When 0, the System Tick interrupt is disabled. When enabled, the interrupt is generated when the System Tick counter counts down to 0.	0
2	CLKSOURCE	System Tick clock source selection. When 1, the CPU clock is selected. When 0, the external clock pin (STCLK) is selected.	1

Table 440. System Timer Control and status register (STCTRL - 0xE000 E010) bit description *…continued*

Bit	Symbol	Description	Reset value
15:3	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
16	COUNTFLAG	System Tick counter flag. This flag is set when the System Tick counter counts down to 0, and is cleared by reading this register.	0
31:17	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

23.5.2 System Timer Reload value register (STRELOAD - 0xE000 E014)

The STRELOAD register is set to the value that will be loaded into the System Tick Timer whenever it counts down to zero. This register is loaded by software as part of timer initialization. The STCALIB register may be read and used as the value for STRELOAD if the CPU or external clock is running at the frequency intended for use with the STCALIB value.

Table 441. System Timer Reload value register (STRELOAD - 0xE000 E014) bit description

Bit	Symbol	Description	Reset value
23:0	RELOAD	This is the value that is loaded into the System Tick counter when it counts down to 0.	0
31:24	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

23.5.3 System Timer Current value register (STCURR - 0xE000 E018)

The STCURR register returns the current count from the System Tick counter when it is read by software.

Table 442. System Timer Current value register (STCURR - 0xE000 E018) bit description

Bit	Symbol	Description	Reset value
23:0	CURRENT	Reading this register returns the current value of the System Tick counter. Writing any value clears the System Tick counter and the COUNTFLAG bit in STCTRL.	0
31:24	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

23.5.4 System Timer Calibration value register (STCALIB - 0xE000 E01C)

The STCALIB register contains a value that is initialized by the Boot Code to a factory programmed value that is appropriate for generating an interrupt every 10 milliseconds if the System Tick Timer is clocked at a frequency of 100 MHz. This is the intended use of the System Tick Timer by ARM. It can be used to generate interrupts at other frequencies by selecting the correct reload value.

Table 443. System Timer Calibration value register (STCALIB - 0xE000 E01C) bit description

Bit	Symbol	Value	Description	Reset value
23:0	TENMS		Reload value to get a 10 millisecond System Tick underflow rate when running at 100 MHz. This value initialized at reset with a factory supplied value selected for the LPC176x/5x. The provided values of TENMS, SKEW, and NOREF are applicable only when using a CPU clock or external STCLK source of 100 MHz.	0x0F 423F
29:24	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	SKEW		Indicates whether the TENMS value will generate a precise 10 millisecond time, or an approximation. This bit is initialized at reset with a factory supplied value selected for the LPC176x/5x. See the description of TENMS above. When 0, the value of TENMS is considered to be precise. When 1, the value of TENMS is not considered to be precise.	0
31	NOREF		Indicates whether an external reference clock is available. This bit is initialized at reset with a factory supplied value selected for the LPC176x/5x. See the description of TENMS above. When 0, a separate reference clock is available. When 1, a separate reference clock is not available.	0

23.6 Example timer calculations

The following examples illustrate selecting System Tick Timer values for different system configurations. All of the examples calculate an interrupt interval of 10 milliseconds, as the System Tick Timer is intended to be used.

Example 1)

This example is for the System Tick Timer running from the CPU clock (cclk), which is 100 MHz.

STCTRL = 7. This enables the timer and its interrupt, and selects cclk as the clock source.

RELOAD = (cclk / 100) - 1 = 1,000,000 - 1 = 999,999 = 0xF423F

In this case, there is no rounding error, so the result is as accurate as cclk.

Example 2)

This example is for the System Tick Timer running from the CPU clock (cclk), which is 80 MHz.

STCTRL = 7. This enables the timer and its interrupt, and selects cclk as the clock source.

RELOAD = (cclk / 100) - 1 = 800,000 - 1 = 799,999 = 0xC34FF

In this case, there is no rounding error, so the result is as accurate as cclk.

Example 3)

This example is for the CPU clock (cclk) is taken from the Internal RC Oscillator (IRC), factory trimmed to 4 MHz.

STCTRL = 7. This enables the timer and its interrupt, and selects cclk as the clock source.

RELOAD = (F_{IRC} / 100) - 1 = 40,000 - 1 = 39,999 = 0x9C3F

In this case, there is no rounding error, so the result is as accurate as the IRC.

Example 4)

This example is for the System Tick Timer running from an external clock source (the STCLK pin), which in this case happens to be 32.768 kHz.

STCTRL = 3. This enables the timer and its interrupt, and selects the STCLK pin as the clock source. STCLK must be selected as the function of the relevant pin. See Section 8.5.6.

RELOAD = (cclk / 100) - 1 = 327.6 - 1 = 327 (rounded up) = 0x0147

In this case, there is rounding error, so the interrupt rate will drift slightly relative to the input frequency.